ECE 340: Semiconductor Devices

Lecture 1: Course Introduction

Prof. Hyunseok Kim

Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

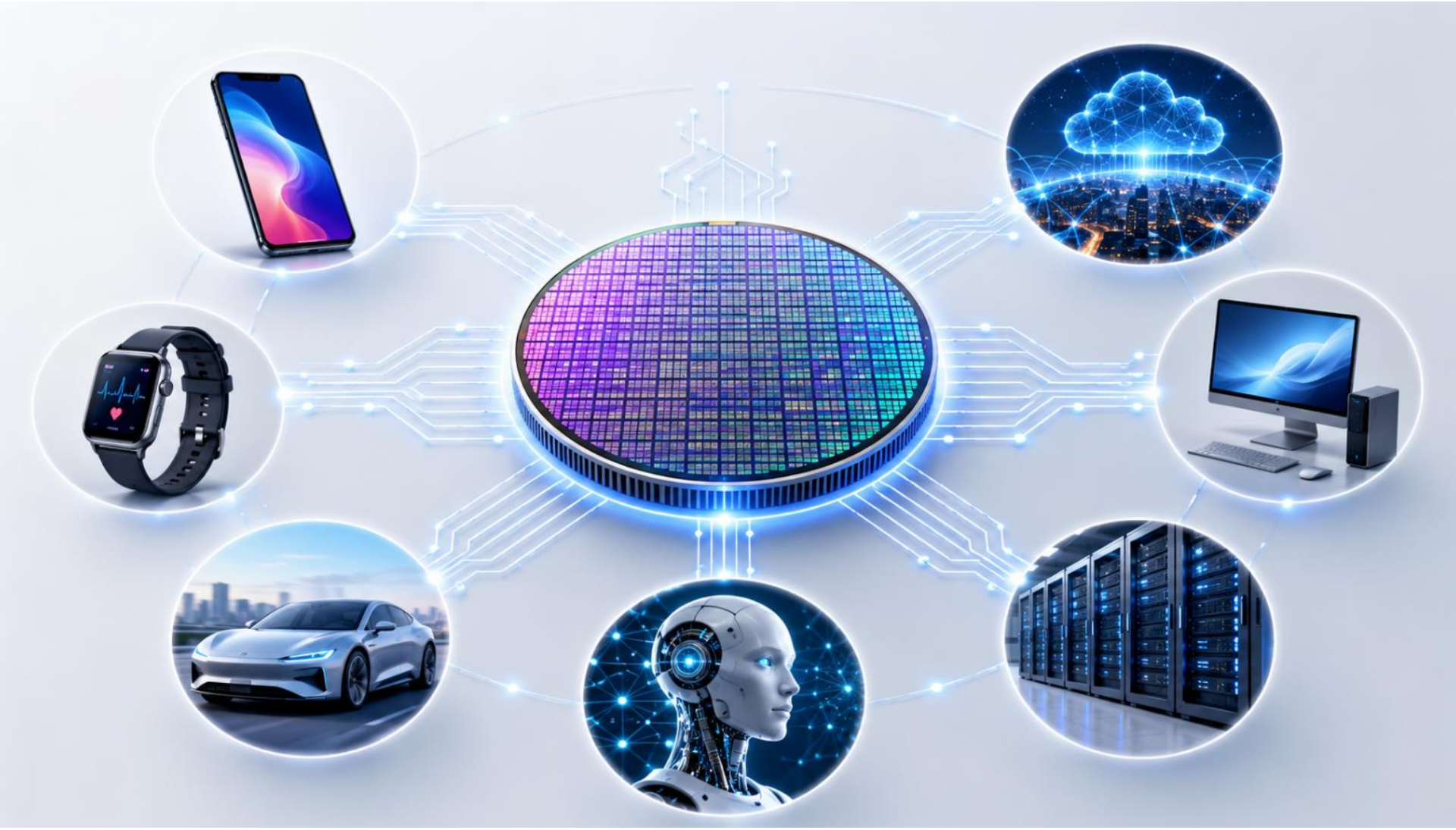
E-mail: hyunseok@illinois.edu

Office Hours: Thursdays 12p-1p, 2256 HMNTL

Course Purpose & Objectives

- Introduce key concepts in **semiconductor materials**
- Provide a basic understanding of **semiconductor devices:**
  - **p-n junctions**
  - **light-emitting diodes (LEDs) and photodetectors (PDs)**
  - **field effect transistors (FETs)**
  - **bipolar junction transistors (BJTs)**

Why is this course important?



Schedule & Policies

Important Information

- Course Website:
  - http://courses.engr.illinois.edu/ece340/
- Download and Review Syllabus / Course Information from Website!
- Canvas: Homework Submission, Exams, & Grades
- Course Director: Prof. John Dallesasse
  - jdallesa@illinois.edu
  - Coordinates schedule, policies, absence issues, homework, quizzes, exams, etc.
- Contact Information and Office Hours for All ECE340 Professors & TAs in Syllabus
- Lecture Slides: Previous Semester Posted on Website
- DRES Students: Contact Prof. Dallesasse ASAP
- Textbook:
  - "Solid State Electronic Devices," Streetman & Banerjee, 7th Edition
  - Supplemental: "Semiconductor Device Fundamentals," Pierret
  - Additional reference texts listed in syllabus

Exams

- Exams In Person, On Campus, Combined, Schedule Announced
- Exam I: Thursday October 15th, 7:00-8:00 pm
- Exam II: Thursday November 17th, 7:00-8:00 pm
- Final Exam: Date/Time To Be Announced
  - Determined by University Registrar & CBTF
- No Cheating
  - Penalties are severe and will be enforced

Homework

- Assigned Friday, Due Following Friday (Typically)
  - Due dates shown in syllabus
- Due at Time Indicated on Canvas
- Follow Guidelines in Syllabus
- Peer Discussions Related to Homework are Acceptable and Encouraged
- Directly Copying Someone Else's Homework is Not Acceptable
  - Graders have been instructed to watch for evidence of plagiarism
  - Both parties will receive a "0" on the problem or assignment

Homework

- No Late Homework
  - Homework due on the date of an excused absence must be turned in ahead of time
  - You must turn in homework in your section
  - No excused absences for homework assignments
  - Top 9 of 11 homework assignments used in calculation of course grade
    - Do all of them to best prepare for the exams!

Absences

- The absence policy in the syllabus will be strictly enforced
- No excused absences are given for homework, to address any situation that might cause you to miss a homework only the best 9 of 11 are used to calculate your final grade
- Excused absences are not given for exams, except in accordance with the UIUC Student Code
- Unexcused work will receive a “0”
- Failure to take the final will result in an “incomplete” grade (if excused) or a “0” (if unexcused)

Grading

Grading Criterion

Homework	10 %
Exam I	25 %
Exam II	25 %
Final Exam	40 %
Total	100 %

Historical Grade Trends*

	Fall 2021	Fall 2023	Fall 2025
A's	24 %	30 %	26 %
B's	30 %	29 %	31 %
C's	26 %	26 %	30%
D's	12 %	9 %	10 %
F's	8 %	5 %	4 %

*Past performance is not necessarily indicative of future results

Assignments

Assignments

- Read info packet – key course policies and schedule are outlined here, including hourly exam dates
- Homework assigned every Friday, due following Friday (HW1 Due on 9/4)
- Begin to read Chapter 1 of Streetman's book
  - Sections 1.1, 1.2, 1.3.1, 1.4
  - I suggest reading all of Chapter 1, but only the above sections are assigned
- Chapter 1 in Pierret covers similar material, and complements Streetman for another perspective

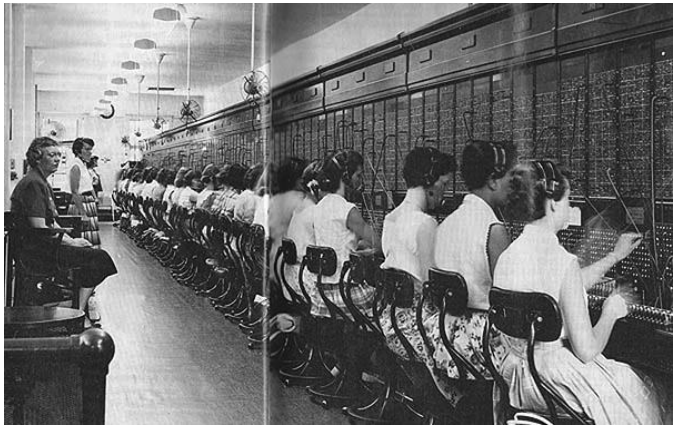
Semiconductor Impact, Historical Perspective, & Illinois Influence

Life Before the Transistor



Calculator





Switch/Router



Television



Cell Phone



Telephone

Electron Tube Shortcomings

- Vacuum Tube Triode
  - Hot cathode needs 5 minutes to warm-up
  - 15 minutes to turn on Radio, TV & radar
  - 120 minutes to turn computer-on
  - Bad Reliability
  - Not amenable to complex systems or integration
  - For a simple ~2,000,000 transistor microprocessor, a “chip” made out of vacuum tubes would be about 300’ X 300’ (not possible)
    - Assuming a 1”X1” tube on a 3” pitch to allow for routing traces, etc.

Early Semiconductor Devices

- 1874: Ferdinand Braun observes rectification in point contact pressed onto lead sulfide (galena), and other materials
- The phenomenon was not understood
- ~1899 Braun patents use of crystal rectifiers in wireless receivers
- 1909 Braun shares Nobel prize with Marconi “in recognition of their contributions to the development of wireless telegraphy”
- Silicon crystals with tungsten whisker contacts became widely used for rectifiers



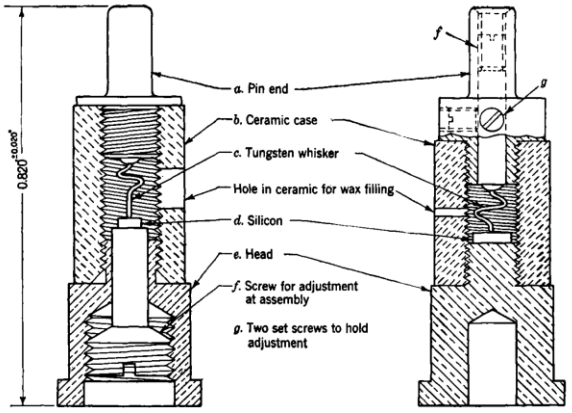


Fig. 2-2.—Ceramic cartridges.

From Phenomenon to Physics

- Postulation of electron wave nature (de Broglie & Schrodinger) and experimental demonstration of wave properties of electrons (Davisson & Germer)
- Sommerfeld proposes a simplified model of a metal – sea of electrons following Fermi-Dirac statistics in a positive metal ion background
- Mott and Schottky develop theories for rectification at metal-semiconductor contacts

Zur Elektronentheorie der Metalle auf Grund der Fermischen Statistik.

I. Teil: Allgemeines, Strömungs- und Austrittsvorgänge.

Von **A. Sommerfeld** in München.

Mit 4 Abbildungen. (Eingegangen am 17. Dezember 1927.)

The theory of crystal rectifiers

By N. F. Mott, F.R.S.

H. H. Wills Physical Laboratory, University of Bristol

(*Received* 24 *January* 1939)

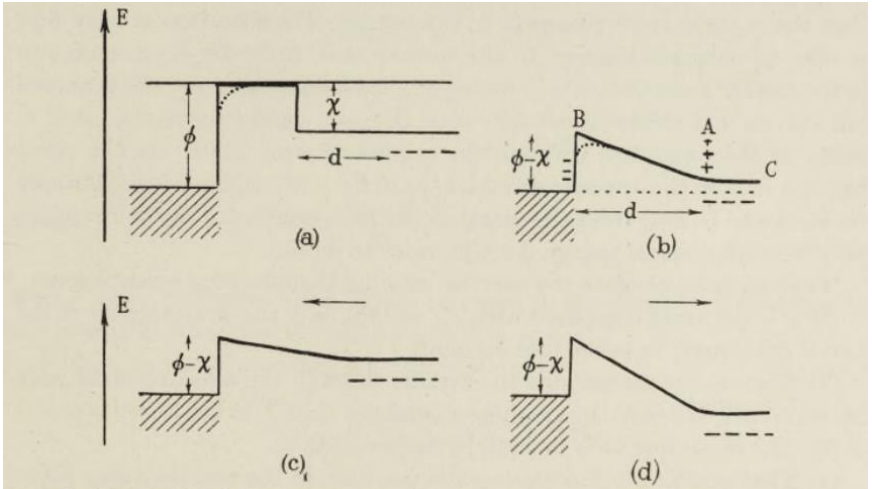
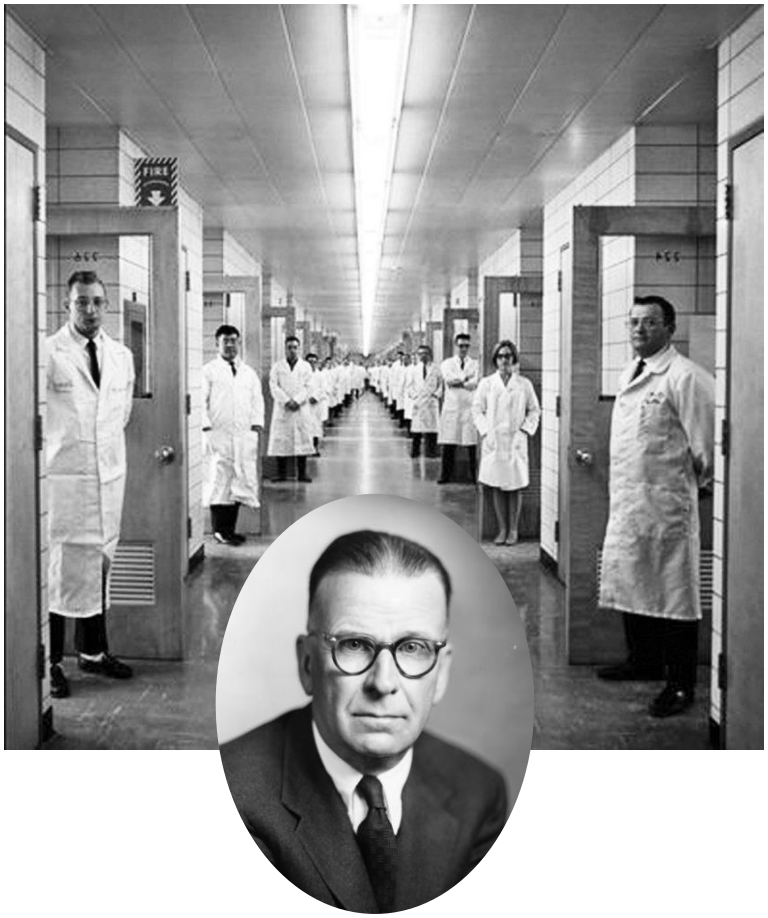


Fig. 3. Energy levels in an impurity semi-conductor separated from a metal by a layer of pure material of thickness *d*. (*a*) See text. (*b*) In equilibrium. (*c*) With a field in the direction of easy flow. (*d*) With a field in the direction of high resistance. The dotted line in (*a*) and (*b*) represents the effect of the image force. The arrows mark the direction of flow of electrons.

Setting the Stage

- 1938: Mervin Kelly (Bell Labs Director of Research) forms group focused on solid state physics, towards the goal of making solid state switches for the telephone system
- 1939: Shockley envisions a semiconductor amplifier, attempts to fabricate using a copper grid with no success
- 1945: Shockley generates another design that fails to function
- Shockley turns to other projects



Who Are These Guys?





Bardeen and Brattain





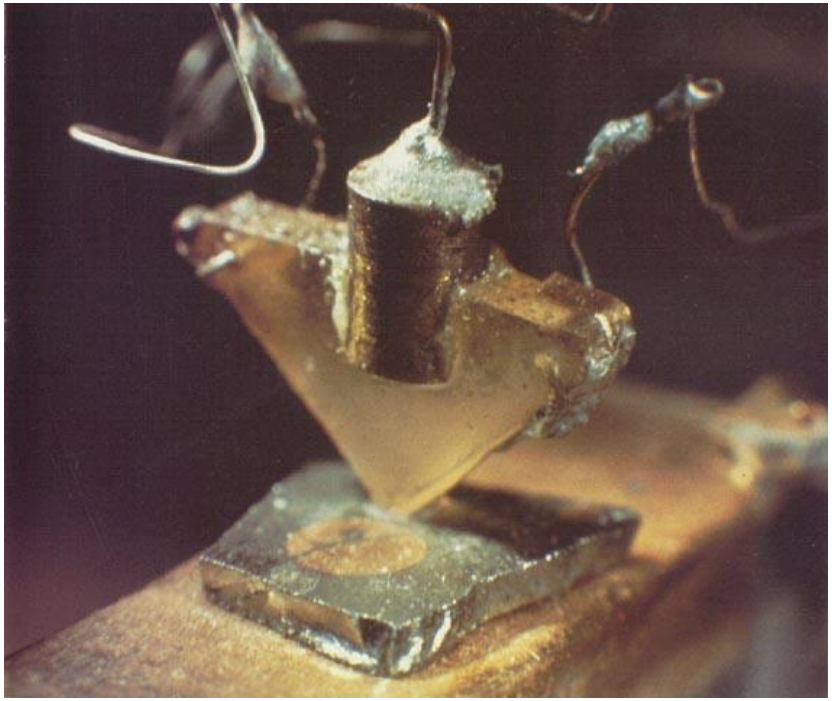
Bardeen and Brattain



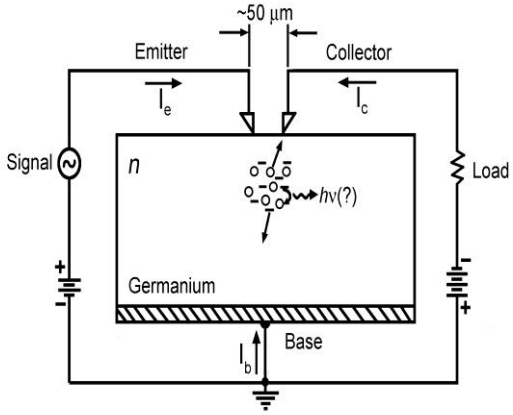


- **Walter Brattain**
  - 1929 (age 27) PhD in Physics from University of Minnesota ("Efficiency of Excitation by Electron Impact and Anomalous Scattering in Mercury Vapor")
  - 1928-1929 National Bureau of Standards, Radio Engineer
  - 1929 Joins Bell Labs, working on copper oxide rectifiers
- **John Bardeen**
  - 1935 (age 27) PhD in Mathematics from Princeton (on Quantum Theory of Solids)
  - 1935-1938 Harvard University Junior Fellow
  - 1938-1941 University of Minnesota
  - 1941-1945 (WWII) Naval Ordnance Lab, Influence Fields (protect from magnetic mines & torpedoes)
  - 1945 Joins Bell Labs

Discovery of Transistor by Bardeen and Brattain (1947)







Identification of minority carrier injection, recombination in the base and collection: underlying bipolar transistor operation

First Commercialization



Manufactured point-contact transistor



Jack A. Morton

- Dec 23, 1947 (1 week after discovery): **Point-contact transistor demonstrated** to Bell Labs executives in circuit amplifying speech into headphones
  - **Needs a name**: John Pierce (friend of Brattain & science fiction writer) combines "transfer" and "resistor"
  - **Shockley blocks Bardeen** from further involvement in transistor work
- June 1948: Jack A. Morton assigned to **lead mass production** of point-contact transistors ("Type A" transistor)
  - "We shall change the world, in what manner I do not know, but change it we will."
- 1949-1950: Transistor **mass production** begins
- July 24, 1951: **Bardeen leaves** Bell Labs for the University of Illinois Urbana-Champaign

Changing the World

- 1947: Invention of the Transistor
- 1957: Theory of Superconductivity



A Few Milestones in Semiconductor Devices and Integrated Circuits (Illinois ECE Connection)

1947: First Bipolar Transistor (Bardeen & Brattain)
1958: Invention of Integrated Circuits (Kilby & Noyce)
1960: First MOSFET (Kahng and Atalla)
1962: First LED and Semiconductor Laser (Hall, Holonyak)
1963: Invention of CMOS (Wanlass and Sah)
1968: Invention of DRAM (Dennard)
1971: First Microprocessor (Intel)



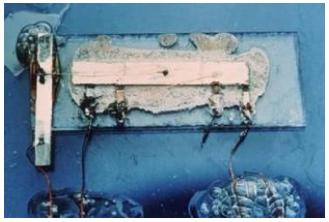
Holonyak: UIUC ECE
B.S. '50, M.S. '51, Ph.D. '54
ECE Professor: 1963-2013

Bardeen:
ECE Professor: 1951-1975



Sah: UIUC ECE B.S. '53
ECE Professor: 1962-1988

Kilby & Noyce: Integrated Circuit, 1958



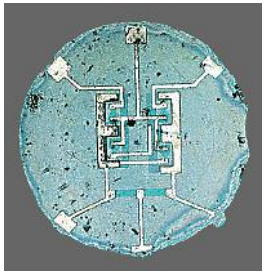
First transistor (1958)
10 mm x 1.5 mm



Jack Kilby

UIUC ECE
B.S. ‘47

Nobel Prize in Physics 2000
“For the invention of integrated circuit”

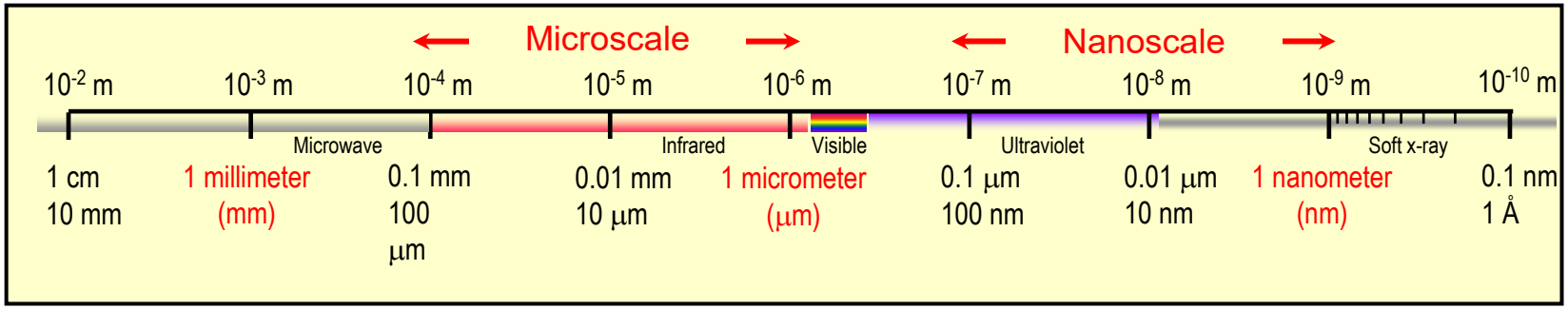




Bob Noyce

Independently developed IC
a few months after Kilby

How small are today's transistors?

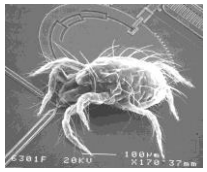




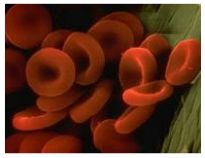
Ant
~ 5 mm



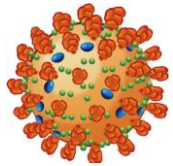
Human hair
~ 100 μm thick



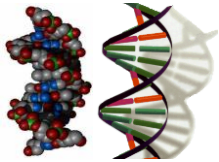
Dust mite
200 μm



Red blood cells
~ 7 μm



SARS-Cov-2
~ 120 nm

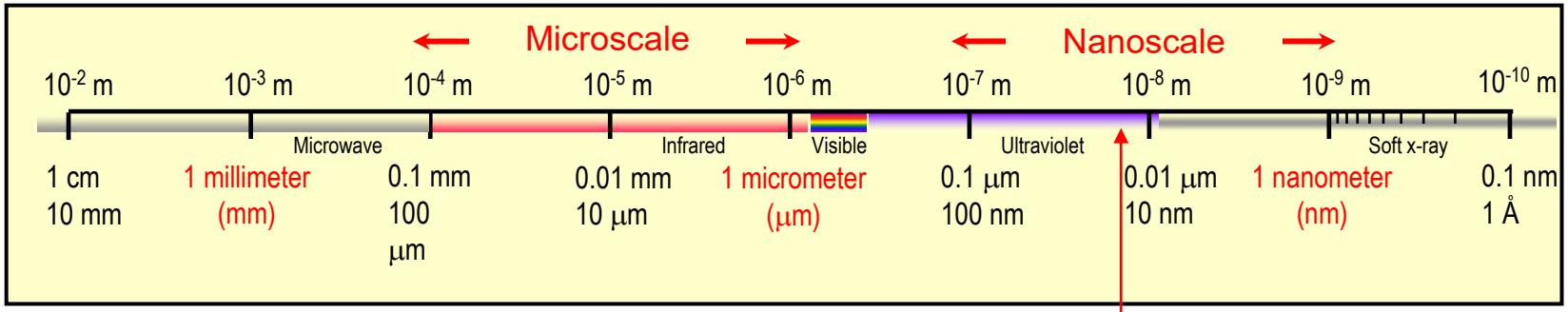


DNA
~2 nm diameter



Atoms of silicon
spacing ~ 5 Å (0.5nm)

How small are today's transistors?



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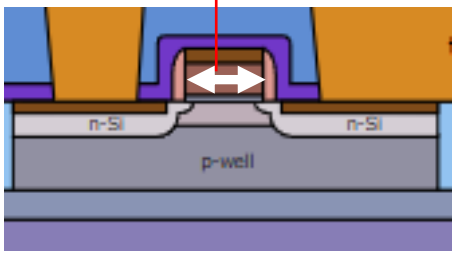
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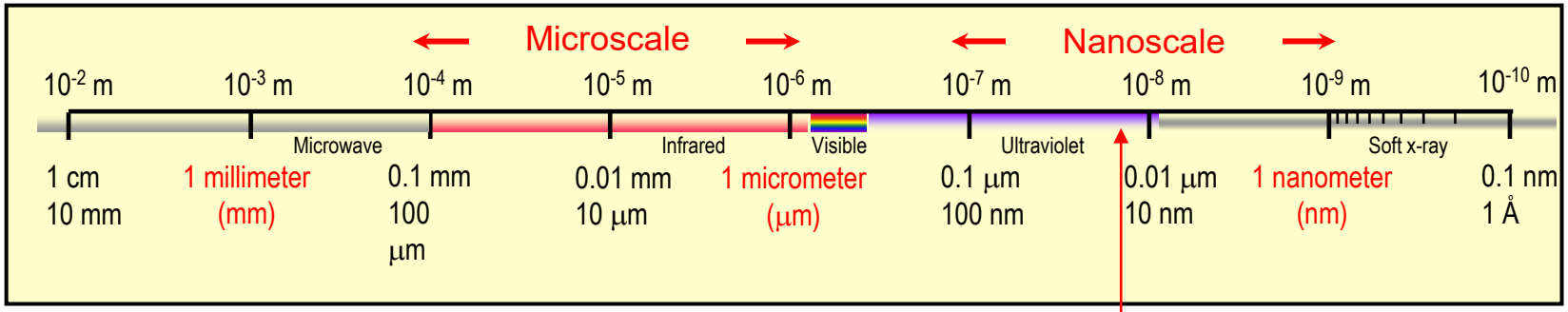
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Transistor gate length (2024)
~ 15 nm

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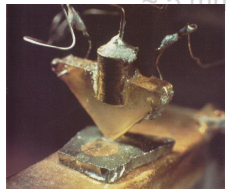




Vacuum tube (1904-)
~ 10 cm

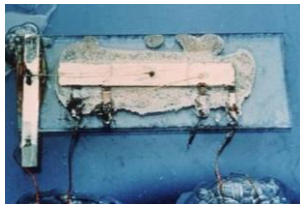
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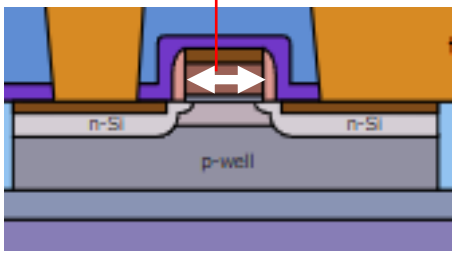
First transistor on IC (1958)
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Red blood cells
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SARS-Cov-2
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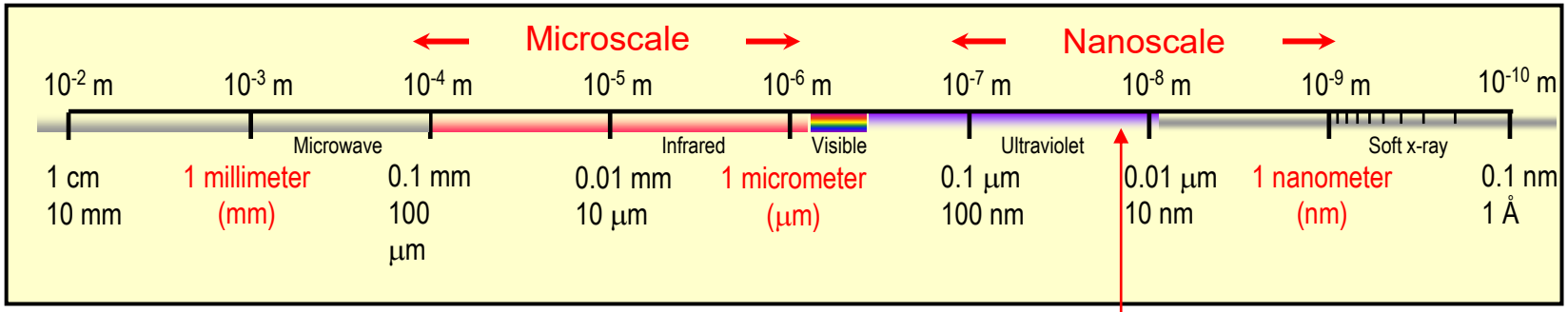
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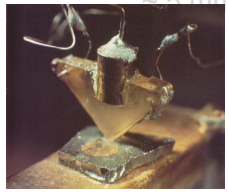




Vacuum tube (1904-)
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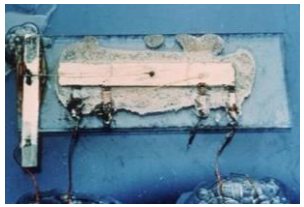
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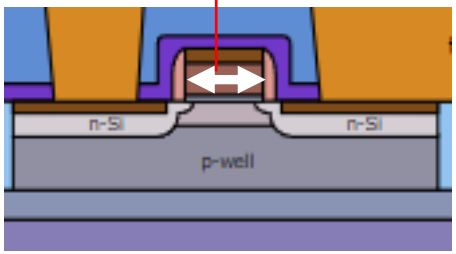
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Transistor gate length (2024)
~ 15 nm

First Practical LED: 1962



Nick Holonyak: Major Awards

National Medal of Science



National Medal of Technology



Global Energy Prize



Draper Prize



Japan Prize



Queen Elizabeth Prize



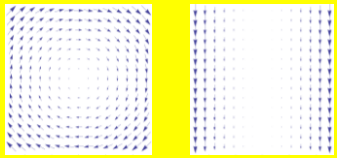
General Introduction to Semiconductor Electronics

Basic Equations for Semiconductor Device Operation

- The basic equations for semiconductor device operation describe the static and dynamic behavior of carriers in semiconductors under the influence of external fields that cause deviation from the thermal equilibrium conditions
- The basic equations can be classified in four groups:
  - Maxwell's Equations
  - Current-Density Equations
  - Continuity Equations
  - Quantum Mechanics & Schrödinger's Equation

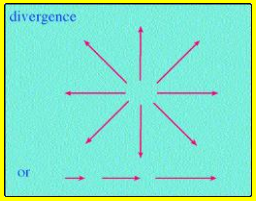
Maxwell Equations for Homogeneous and Isotropic Materials

$$\nabla \times E = -\frac{\partial B}{\partial t}$$



$$\nabla \times H = -\frac{\partial D}{\partial t} + J_{cond} = J_{total}$$

$$\nabla \cdot D = \rho(x,y,z)$$



$$\nabla \cdot B = 0$$

$$B = \mu_o H$$

$$D(r,t) = \int_{-\infty}^{t} \varepsilon_s(t-t')E(r,t')dt'$$

$$= \varepsilon E + P$$

E: electric field

D: electric displacement

B: magnetic field

H: magnetizing field

ε_s: permittivity

μ_o: permeability

ρ: total electric charge density

J_{cond}: the conduction current density

P: polarization density

$\times$: curl operator

$\bullet$: divergence operator

Key Points: Fields & Waves

Charge creates divergence in the electric displacement field:

$$\nabla \cdot D = \rho$$

The electric field is related to the electric displacement field through the permittivity:

$$D = \varepsilon \mathcal{E}$$

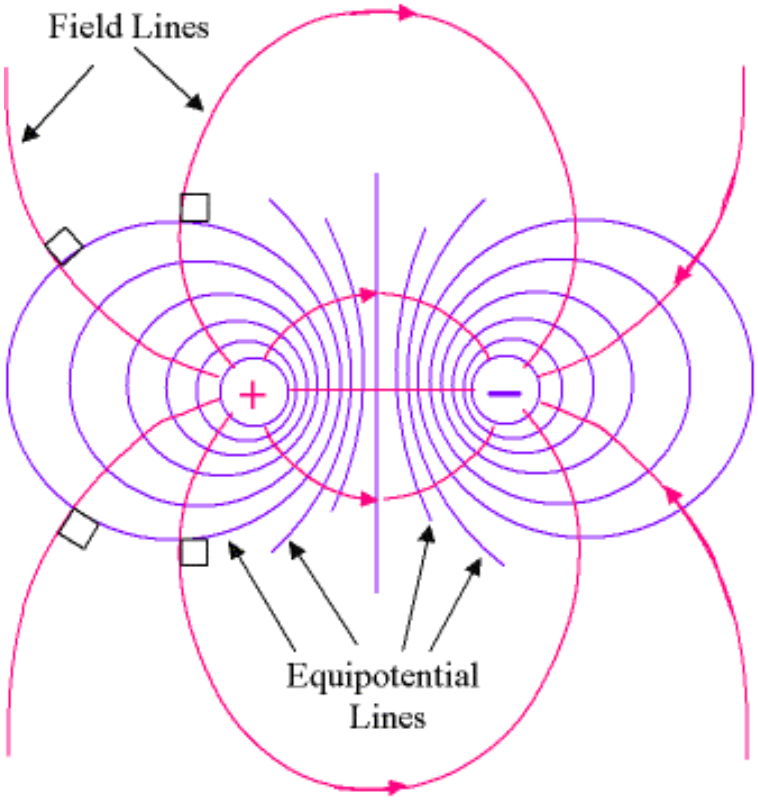
$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}$$

The electric field is the negative of the gradient of the scalar potential field:

$$\mathcal{E} = -\nabla \mathcal{V}$$

In one dimesion: $\mathcal{E}(x) = -\dfrac{d\mathcal{V}(x)}{dx}$

$$\mathcal{V}(x) = -\int \mathcal{E}(x)\,dx$$



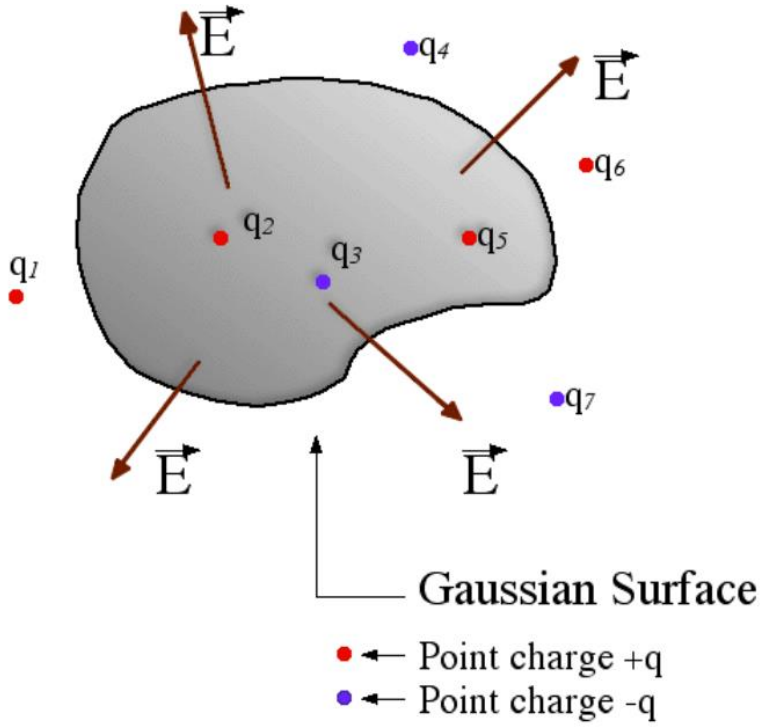
http://www.pstcc.edu/departments/natural_behavioral_sciences/Web%20Physics/E2020D0103.gif

Gauss' Law

$$\nabla \cdot \mathbf{D} = \rho$$

ρ is the charge density with units of $\frac{Coulombs}{m^3}$

- The divergence of the electric displacement flux density is equal to the charge density



http://www.ibiblio.org/links/devmodules/Gauss/graphics/blob1.gif